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Abstract

Cybersecurity threats are growing every year, both in number and in how advanced they are. Traditional rule-based and signature-based security tools are struggling to keep up, especially against new or previously unseen (zero-day) attacks. This report reviews recent academic literature (2020-2024) on how artificial intelligence (AI) and machine learning (ML) techniques are being used to improve the efficiency and effectiveness of cybersecurity threat detection systems, compared to traditional approaches. Twenty peer-reviewed sources were selected from IEEE Xplore, ACM Digital Library, ScienceDirect, MDPI, and Springer, covering network intrusion detection, malware and ransomware detection, phishing detection, DDoS detection, IoT security, and the use of generative AI in cybersecurity. The review finds that ML and deep learning (DL) models generally outperform traditional methods in detection accuracy and adaptability, but they also come with real limitations, including vulnerability to adversarial attacks, heavy reliance on the quality of training data, and a lack of transparency in how decisions are made. The report also highlights gaps in current research, particularly around real-world testing and the dual-use risk of generative AI tools, which shape the direction of this ongoing study.

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1. Introduction & Background

Cybersecurity has become one of the biggest challenges for organisations and individuals in the digital world. Every year, cyberattacks increase in number, and attackers keep finding new ways to bypass existing security controls. For a long time, cybersecurity systems mainly relied on rule-based or signature-based detection, where a system can only detect an attack if it matches a known pattern that has already been recorded. This worked reasonably well when threats were simple and predictable, but it is no longer good enough, since modern attacks are constantly evolving and many, especially zero-day attacks, do not match any existing signature.

Because of this, researchers have increasingly turned to artificial intelligence (AI) and machine learning (ML) to build smarter, more adaptive threat detection systems. Instead of relying only on fixed rules, ML and deep learning (DL) models can learn from large amounts of data and recognise patterns that a human analyst or a static rule might miss. These techniques have been applied across many areas of cybersecurity, including network intrusion detection, malware detection, phishing detection, and DDoS attack detection.

This literature review focuses on the research topic assigned in Week 1: the impact of AI and ML on cybersecurity, specifically how AI/ML techniques affect the efficiency and effectiveness of threat detection systems compared to traditional, rule-based approaches. The review brings together twenty recent academic sources published between 2020 and 2024 to understand what has already been discovered, what methods and datasets researchers are using, and where the gaps in current knowledge still exist. This background sets up the research question and hypothesis of this study, which propose that AI/ML techniques significantly improve the efficiency and accuracy of cybersecurity threat detection compared to traditional methods.

2. Purpose and Aims of the Literature Review

The purpose of this literature review is to build a clear picture of what is already known about AI/ML-based cybersecurity threat detection before carrying out further research on the topic. More specifically, this review aims to:

- Establish what is currently known about how effective AI/ML techniques are at detecting cyberattacks, particularly zero-day and previously unseen attacks, compared to traditional rule-based methods.
- Identify the most common models, datasets, and evaluation methods that researchers use in this field, so that this can inform the design of future research.
- Identify important research gaps that justify why this research question and hypothesis are worth investigating further.
- Present a balanced view of AI/ML in cybersecurity that considers both the benefits (higher accuracy, adaptability) and the risks/limitations (adversarial vulnerability, dataset dependence, dual-use of generative AI).

3. Literature Review Process

A structured literature search was carried out across five major academic databases: IEEE Xplore Digital Library, ACM Digital Library, ScienceDirect, Springer Digital Library, and MDPI (mainly the Sensors and Electronics journals), with Google Scholar used to cross-check citation counts and locate full text where needed. The keyword combinations used included: “artificial intelligence AND cybersecurity”, “machine learning AND intrusion detection”, “deep learning AND cyber attack detection”, “zero-day attack detection”, “malware detection AND deep learning”, “phishing detection AND machine learning”, “adversarial machine learning AND cybersecurity”, “federated learning AND intrusion detection”, “ChatGPT OR generative AI AND cybersecurity threat”, and “network intrusion detection system survey”.

To keep the review focused and relevant, the following inclusion criteria were applied: articles had to be published between 2020 and 2024, directly related to the use of AI, ML, or DL in a cybersecurity context, published in a peer-reviewed journal or reputable conference proceeding, available in full text in English, and either cited by other academic work or published in a well-known, indexed outlet (IEEE, ACM, Elsevier, MDPI, Springer, Wiley). Sources were excluded if they were non-peer-reviewed blog posts, vendor marketing material, not available in full text, or not directly related to the cybersecurity domain.

From the initial search results, twenty articles were selected for this review based on their relevance to the research question, the strength of their methodology, and how well they covered different sub-domains of cybersecurity, including network intrusion detection, malware and ransomware detection, phishing detection, DDoS detection, IoT security, adversarial robustness, and the emerging role of generative AI. Each article was read in full, and the key findings, methods used, and limitations were recorded and compared, as summarised in the sections below.

4. Main Body of the Report

4.1 Machine Learning and Deep Learning in Network Intrusion Detection

Network intrusion detection systems (NIDS) are one of the most heavily researched areas of AI-based cybersecurity. Ozkan-Okay et al. [1] carried out a broad survey comparing supervised, unsupervised, deep learning, and reinforcement learning approaches, and concluded that no single technique works best for every attack type, and that hybrid models tend to achieve better detection rates. Momand, Jan and Ramzan [2] and Pinto et al. [3] both carried out systematic reviews of ML/DL-based intrusion detection systems, testing them against common benchmark datasets such as NSL-KDD, CICIDS2017, and UNSW-NB15. Ali et al. [4] added to this by reviewing ML strategies used specifically in intrusion detection, again confirming that ML-based systems consistently outperform static, rule-based detection when identifying new attack types.

Ahmad et al. [5] carried out one of the earlier and most cited systematic studies in this space, and found that deep learning models are increasingly preferred because they can automatically learn important features from raw traffic data instead of relying on manual feature engineering. More recently, Abdulganiyu, Ait Tchakoucht and Saheed [6] conducted an updated review of NIDS research, noting that while detection accuracy keeps improving on paper, there remain real concerns about how well these models generalise from lab-based datasets to live, real-world network traffic.

4.2 Malware and Ransomware Detection

A second major theme is the use of deep learning for malware detection. Bensaoud, Kalita and Bensaoud [7] and Deldar and Abadi [8] both concluded that DL models generalise noticeably better to previously unseen (zero-day) malware than older, signature-based antivirus tools, although this comes at the cost of needing larger, more diverse training datasets. This theme continues into ransomware research: Alraizza and Algarni [9] found that ML-based approaches consistently outperform traditional signature-based ransomware detection, particularly for unseen ransomware families, while stressing how important a large, representative training dataset is. Ispahany et al. [10] built on this with a more recent review, noting that many studies still test on relatively small or outdated datasets, which limits confidence in reported accuracy figures.

4.3 Phishing Detection Using Machine Learning

Phishing detection is another area where ML has clearly moved beyond traditional methods. Zieni, Massari and Calzarossa [11] and Safi and Singh [12] both reviewed ML-based phishing website detection, reporting high accuracy for supervised classifiers such as Random Forest and Support Vector Machines when trained on URL and webpage features, but both also flagged rapid “concept drift” as an ongoing problem, since phishing tactics change quickly and models trained on older data can quickly become outdated. Atawneh and Aljehani [13] extended this research into phishing emails

specifically, comparing CNN, LSTM, RNN, and BERT-based deep learning models, and found that transformer-based models like BERT achieved very high accuracy, showing that newer deep learning architectures continue to push detection performance higher, though often requiring more computing resources than simpler ML models.

4.4 DDoS Attack Detection

Distributed denial-of-service (DDoS) attacks remain one of the most common and disruptive types of cyberattack, and ML-based detection has become the main research direction here as well. Najafimehr, Zarifzadeh and Mostafavi [14] produced a comprehensive taxonomy of ML-based DDoS detection methods, reviewing supervised, unsupervised, and hybrid approaches. Their review found that combining multiple ML techniques (hybrid models) generally achieves lower false-positive rates and higher detection rates than any single method used alone, which lines up with similar conclusions drawn in the intrusion detection literature above.

4.5 IoT Security and Federated Learning Approaches

As more devices connect to the internet, securing the Internet of Things (IoT) has become its own major research area. Al-Garadi et al. [15] carried out a widely cited survey of ML and DL methods for IoT security, highlighting that IoT devices bring unique challenges, such as limited processing power and diverse device types, which make many “standard” ML-based solutions harder to apply directly. To address privacy and scalability concerns in these distributed environments, Khraisat et al. [16] reviewed the growing use of federated learning (FL) for intrusion detection, where multiple devices train a shared detection model without sending raw data to a central server. They found FL-based intrusion detection promising, especially for privacy-sensitive environments like IoT, but it also introduces new challenges around combining updates from many devices fairly and securely.

4.6 Adversarial Robustness and Limitations of AI-Based Defences

While the literature broadly agrees that AI/ML improves detection performance, almost every source above also raises the same underlying concern: these systems remain vulnerable to adversarial manipulation. Macas, Wu and Fuertes [17] specifically studied how adversarial examples, which are small and deliberately crafted changes to input data, can trick deep learning-based cybersecurity systems into misclassifying malicious traffic as safe. Their review catalogued current defence strategies, such as adversarial training, but noted that no defence currently available is fully reliable against a determined attacker. He, Kim and Asghar [18] carried out a similarly detailed survey focused specifically on adversarial machine learning in network intrusion detection, and reached a similar conclusion: as deep neural networks are used more widely in NIDS because of their high accuracy, they simultaneously become a bigger target for adversarial attacks, creating a kind of trade-off between accuracy and robustness that current research has not fully solved.

4.7 Generative AI and Large Language Models: A Dual-Use Risk

A newer and rapidly growing theme in the literature is the dual-use nature of generative AI and large language models (LLMs) in cybersecurity. Chowdhury et al. [19] examined how tools like ChatGPT can be genuinely useful for defenders, for example by helping analysts write or explain code, while also being misused by attackers to threaten the confidentiality, integrity, and availability (the “CIA triad”) of systems, for instance by helping to write convincing phishing messages or basic malicious scripts. Chen et al. [20] built on this with a broader survey of how LLMs are being used across the full cyber threat detection pipeline, from defensive reconnaissance to identifying attacker behaviour after a system has already been compromised, concluding that while LLMs show real promise as an additional defensive tool, this is still a very new research area with far fewer large-scale, real-world evaluations compared to more established ML/DL methods discussed elsewhere in this review.

4.8 Critical Analysis

Taken together, this body of literature provides fairly strong support for the hypothesis of this study, that AI/ML techniques improve the efficiency and accuracy of cybersecurity threat detection compared to traditional, rule-based approaches. Across every sub-domain reviewed, from network intrusion detection to malware, ransomware, phishing, and DDoS detection, ML and DL models are consistently reported to outperform static, signature-based methods, particularly against zero-day or previously unseen attacks. This is a genuine strength of the literature: the finding is repeated independently across several different research groups, datasets, and attack types, which adds confidence to the overall conclusion.

However, a critical reading also reveals important weaknesses. Many reported accuracy figures come from testing on fixed, publicly available benchmark datasets such as NSL-KDD, CICIDS2017, and UNSW-NB15. These are useful for comparing models fairly, but several studies, including [3] and [6], point out that they do not fully reflect the messiness of real-world, live network traffic, so high accuracy in a lab setting does not always translate into the same performance in practice. Second, the adversarial robustness literature [17], [18] shows that the very technique giving ML/DL its advantage, learning complex patterns from data, is also what makes it exploitable, since attackers who understand roughly how a model works can craft inputs specifically designed to fool it. This means efficiency gains from AI/ML are not unconditional; they depend on the system not being deliberately targeted by an informed attacker. Third, several studies [7], [9], [10] repeatedly identify dataset quality, size, and diversity as the single biggest practical bottleneck, suggesting that progress in this field depends as much on better data collection as on new algorithms.

There are also ethical considerations worth noting. AI in cybersecurity raises privacy questions, particularly when systems need large volumes of network or user data to train effectively; this is part of why federated learning approaches [16] have become more popular, since they aim to improve privacy while still allowing collaborative model training. There is also a clear dual-use dimension to this research: the same generative AI capabilities that help defenders [19], [20] can just as easily be used by attackers, meaning that publishing detailed AI-based defensive techniques risks informing more sophisticated attacks, a tension not yet fully resolved in the literature.

4.9 Research Gaps Identified

Based on this review, several research gaps stand out. Most studies test their models on a single, fixed benchmark dataset rather than validating performance across multiple, more realistic, real-time traffic sources, which limits confidence in how well reported results would hold up in a live deployment. Adversarial robustness remains a consistently unresolved problem across almost every sub-domain reviewed, rather than something that has been “solved” in any one area. Most studies also tend to report efficiency and accuracy improvements for one specific attack type or cybersecurity domain, such as malware or phishing, in isolation, rather than evaluating how AI/ML affects the efficiency and effectiveness of threat detection more broadly across an entire security system, which is exactly the gap that this research question is designed to explore. Finally, the dual-use risk of generative AI and LLMs in cybersecurity is a genuinely new and still under-researched area, with far fewer large-scale studies compared to more established ML/DL techniques, suggesting this will likely remain an important area for future research.

5. Conclusions

This literature review examined twenty recent academic sources published between 2020 and 2024 on the impact of artificial intelligence and machine learning on cybersecurity threat detection. Across network intrusion detection, malware and ransomware detection, phishing detection, DDoS detection, and IoT security, the reviewed literature consistently shows that ML and deep learning-based systems are more accurate and more adaptable than traditional, rule-based approaches, particularly when it comes to identifying new or previously unseen attacks. This provides reasonably strong

support for the hypothesis that AI/ML techniques significantly improve the efficiency and accuracy of cybersecurity threat detection systems compared to traditional methods.

At the same time, the literature makes clear that these gains are not without real limitations. AI/ML-based defences remain vulnerable to adversarial manipulation, their performance depends heavily on the quality and diversity of available training data, and there are unresolved ethical questions around privacy and the dual-use potential of generative AI tools. The emergence of large language models as both a defensive tool and a potential attack enabler is a particularly new and still under-researched area. Overall, this review confirms that AI/ML offers a genuine, measurable improvement over traditional cybersecurity approaches, while also identifying the specific gaps, particularly around real-world validation and adversarial robustness, that this ongoing research aims to help address.

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